

DP Computer Science: A 4.1.1 Types of Machine Learning

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Describe the unique characteristics of **Supervised, Unsupervised, Reinforcement, Deep, and Transfer Learning**
- Differentiate between the various machine learning approaches based on their **learning process** and **data requirements**
- Analyse a complex, real-world problem and propose an appropriate ML approach, justifying the choice with specific reasons
- Identify and describe technical applications for each type of machine learning, such as **sentiment analysis, robotics navigation, and medical imaging diagnostics**

Key Vocabulary

Term	Definition
Machine Learning (ML)	A subset of artificial intelligence that enables systems to learn from data without being explicitly programmed
Supervised Learning	Learning from labelled data; the model learns to map inputs to known outputs
Unsupervised Learning	Learning from unlabelled data; the model finds hidden patterns or groupings
Reinforcement Learning	Learning through trial and error; an agent learns to take actions to maximise rewards
Deep Learning	A subset of ML using neural networks with many layers; excels at pattern recognition (images, audio, text)
Transfer Learning	Using knowledge gained from one task to improve performance on a different but related task
Labelled data	Data that has input-output pairs (e.g., images with known classifications)
Unlabelled data	Data without predefined outputs (e.g., a collection of customer purchase histories)
Agent	In RL, the entity that takes actions in an environment
Reward	In RL, a signal that indicates whether an action was good or bad
Neural network	A computational model inspired by the human brain, consisting of layers of interconnected nodes

Approaches to Learning (ATL) Elements

Skill	Description
Thinking (Critical Thinking)	Analysing complex problems to determine the most suitable ML approach; evaluating characteristics and limitations of different algorithms
Research (Information Literacy)	Gathering, interpreting, and synthesising information on real-world ML applications
Communication	Collaborating in groups; articulating a clear, well-reasoned justification for a chosen ML approach

2. Introduction: The Guiding Question

Big Picture Question

"What principles and approaches should be considered to ensure machine learning models produce accurate results ethically?"

The Hook: How Does Netflix Know What You Want to Watch?

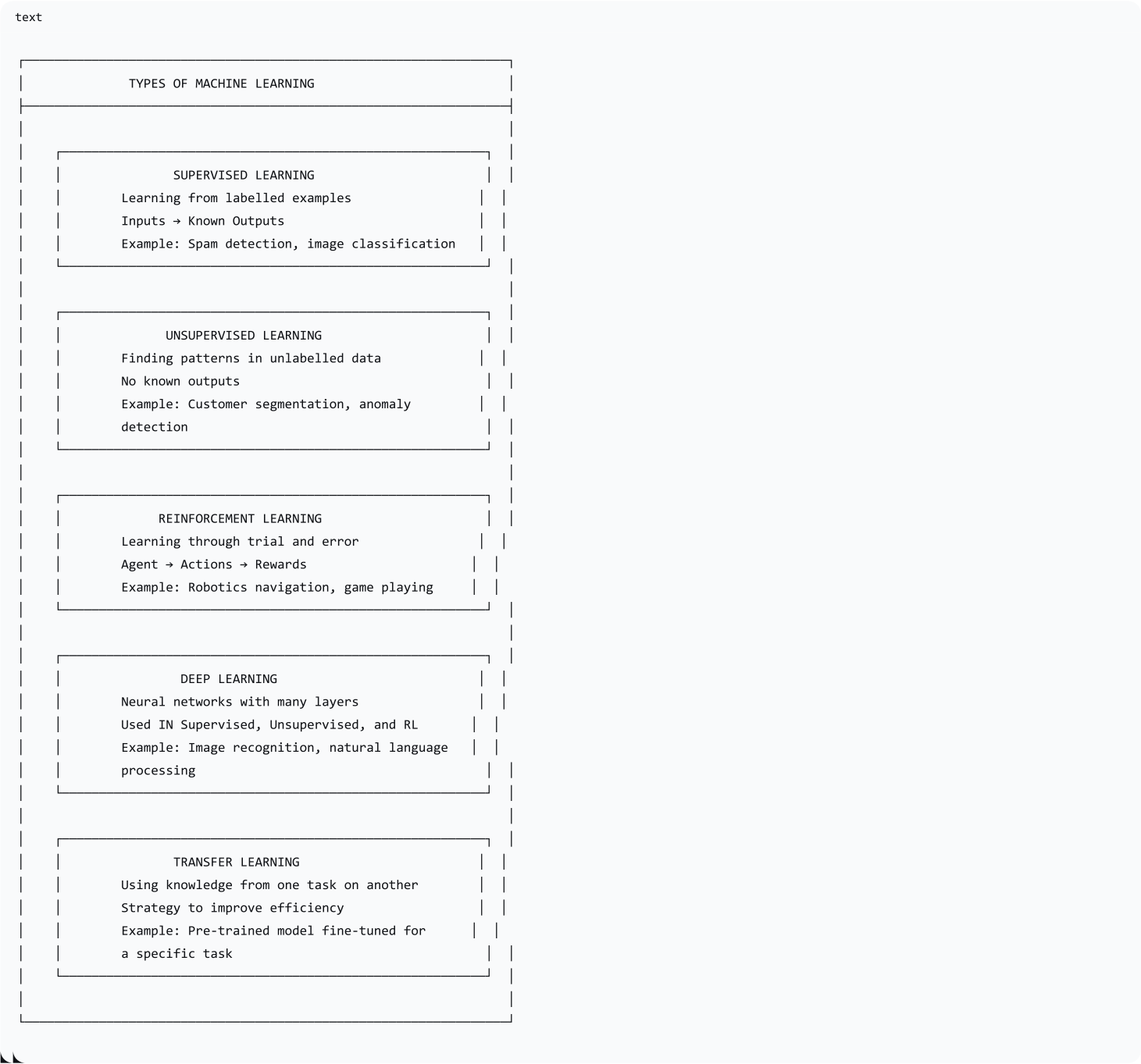


Key Insight

"You've described a process where a system learns from a huge set of existing examples—your past viewing habits (inputs) and your ratings (outputs). In computer science, we call this **Supervised Learning**."

3. The Five Types of Machine Learning

Overview



4. Detailed Explanation of Each Approach

4.1 Supervised Learning

| Supervised Learning involves learning from labelled data—the model learns to map inputs to known outputs.

Aspect	Description
Data Type	Labelled data (input-output pairs)
Goal	Predict outputs for new inputs

Aspect	Description
Common Techniques	Classification, Regression
Analogy	Learning with a teacher who provides correct answers

How It Works:



Technical Applications:

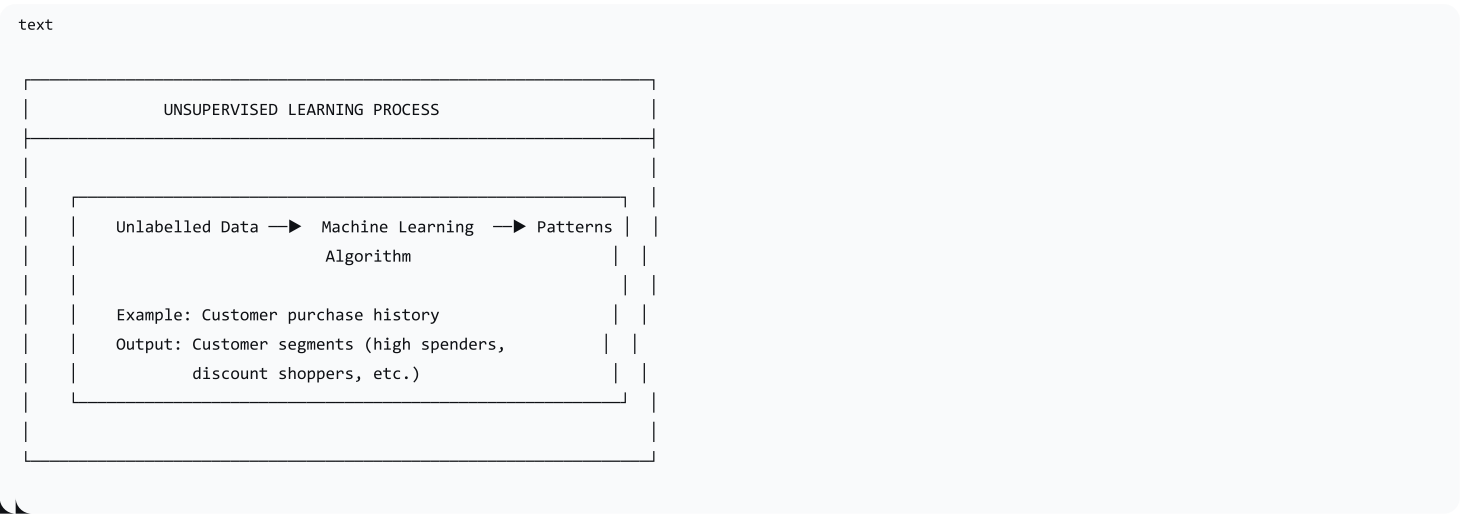
Application	Description
Spam Detection	Classify emails as spam or legitimate
Image Classification	Identify objects in images (cat, dog, car)
Medical Imaging Diagnostics	Detect diseases from X-rays, MRIs
Sentiment Analysis	Determine sentiment (positive/negative) in text
Credit Scoring	Predict loan default risk

4.2 Unsupervised Learning

Unsupervised Learning involves finding patterns in unlabelled data without predefined outputs.

Aspect	Description
Data Type	Unlabelled data
Goal	Discover hidden patterns, groupings, or structure
Common Techniques	Clustering, Dimensionality Reduction, Anomaly Detection
Analogy	Learning without a teacher—discovering patterns independently

How It Works:



Technical Applications:

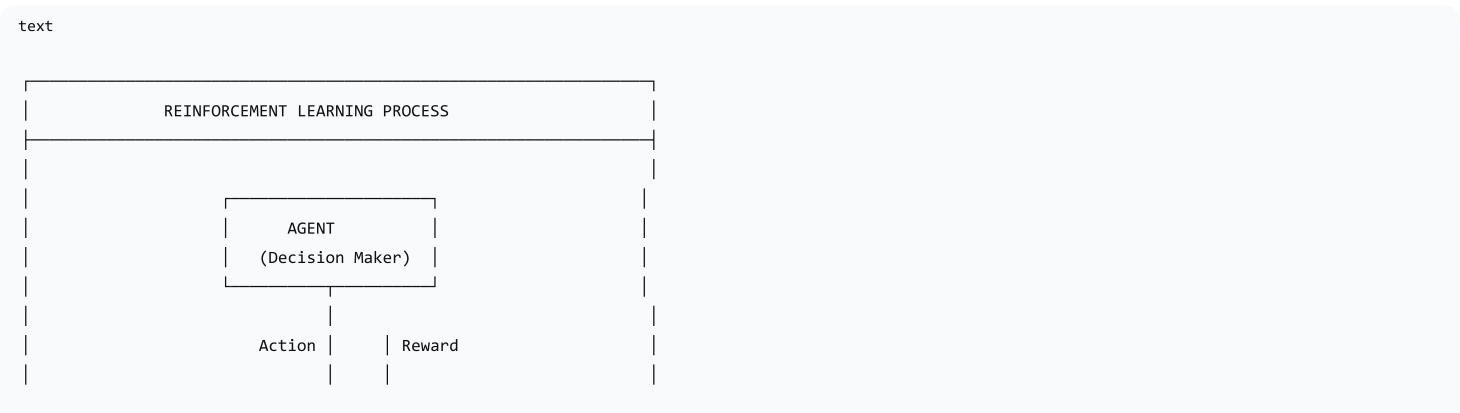
Application	Description
Market Basket Analysis	Identify products frequently bought together
Customer Segmentation	Group customers by behaviour
Anomaly Detection	Find unusual data points (fraud detection)
Recommendation Systems	Find similar users/items for recommendations
Dimensionality Reduction	Simplify data while preserving key information

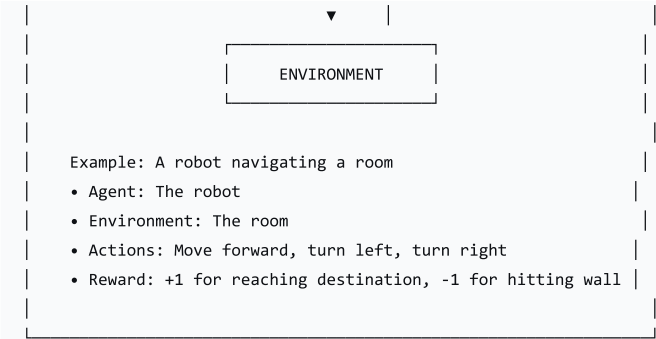
4.3 Reinforcement Learning

Reinforcement Learning involves an agent learning through trial and error in an environment to maximise cumulative rewards.

Aspect	Description
Data Type	Interactive (actions and rewards)
Goal	Learn an optimal policy (strategy) for taking actions
Key Concepts	Agent, Environment, Actions, Rewards, Policy
Analogy	Learning to play chess by playing games and getting feedback (win/lose)

How It Works:





Technical Applications:

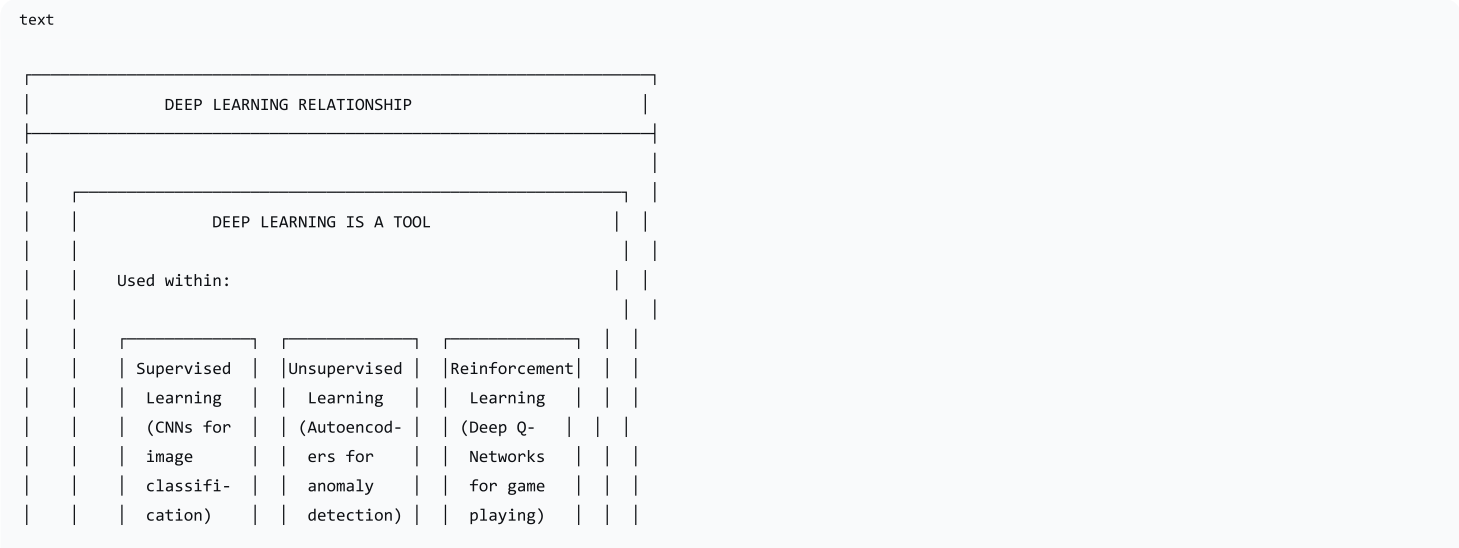
Application	Description
Robotics Navigation	Robots learn to navigate environments
Game Playing	AlphaGo, game-playing AI
Traffic Optimisation	Adaptive traffic light control
Resource Management	Optimising power usage, network traffic
Personalised Learning	Adaptive learning paths for users

4.4 Deep Learning

Deep Learning uses neural networks with many layers to learn representations of data. It is a technique used within supervised, unsupervised, and reinforcement learning.

Aspect	Description
Data Type	Large datasets (can be labelled or unlabelled)
Goal	Learn hierarchical representations from data
Key Concept	Neural networks with multiple hidden layers
Analogy	The brain—layers of neurons processing information

Relationship to Other ML Types:



Deep Learning = Many-layered neural networks

Technical Applications:

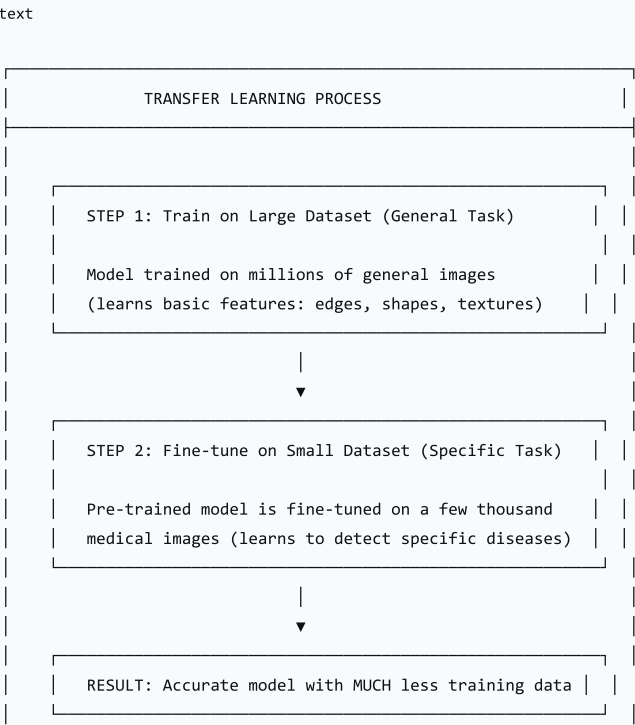
Application	Description
Image Recognition	Object detection, facial recognition
Natural Language Processing	Language translation, text generation
Speech Recognition	Voice assistants (Siri, Alexa)
Autonomous Vehicles	Self-driving car perception
Medical Imaging	Disease detection from scans

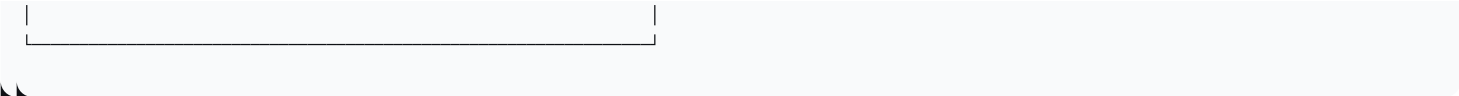
4.5 Transfer Learning

Transfer Learning involves using knowledge gained from one task to improve performance on a different but related task.

Aspect	Description
Data Type	Pre-trained model + task-specific data
Goal	Reduce data requirements and training time
Key Concept	Reusing learned features and representations
Analogy	Learning to ride a bicycle helps you learn to ride a motorcycle

How It Works:





Technical Applications:

Application	Description
Medical Imaging	Pre-trained on general images, fine-tuned for disease detection
Natural Language Processing	Pre-trained language model (BERT), fine-tuned for sentiment analysis
Image Classification	Pre-trained on ImageNet, fine-tuned for specific objects
Robotics	Transfer skills from simulation to real robots

5. Comparison Summary Table

Feature	Supervised	Unsupervised	Reinforcement	Deep Learning	Transfer Lear
Data Type	Labelled	Unlabelled	Interactive	Large datasets	Pre-trained +
Learning Process	Map inputs to outputs	Find hidden patterns	Trial and error	Layer-wise representation	Reuse knowle
Key Technique	Classification, Regression	Clustering, Association	Q-Learning, Policy Gradients	Neural Networks	Fine-tuning
Goal	Predict outputs	Discover structure	Maximise reward	Learn hierarchical features	Transfer know
Example	Spam detection	Customer segmentation	Robot navigation	Facial recognition	Medical imag

6. ML Consultant Project Activity

Scenario A: Optimising City Traffic Flow in Real-Time

Aspect	Analysis
Problem	City traffic lights need to adapt to changing traffic conditions in real-time
Best Approach	Reinforcement Learning
Justification	The problem requires dynamic decision-making in a changing environment. The agent (traffic control system) takes actions (changing light timings), observes the environment (sensors), and receives rewards (reduced congestion).
Why Others Are Less Suitable	Supervised Learning would require labelled data for every possible traffic situation—impossible to create. Unsupervised Learning could identify patterns (rush hour), but cannot take actions to optimise the system.

Scenario B: Early Detection of Crop Disease from Drone Imagery

Aspect	Analysis
Problem	Classify drone images of crops as "healthy" or "diseased"
Best Approach	Supervised Learning with Deep Learning + Transfer Learning
Justification	This is a classic image classification problem. A CNN (Deep Learning) is ideal for image recognition. Transfer Learning allows us to use a pre-trained model on general images and fine-tune it on our specific crop images, reducing data requirements.
Why Others Are Less Suitable	Reinforcement Learning is not applicable (no interactive environment). Unsupervised Learning could group similar images but cannot assign critical "healthy/diseased" labels without prior knowledge.

Scenario C: Personalised Language Learning App

Aspect	Analysis
Problem	An app that adapts to user mistakes to provide personalised learning
Best Approach	Reinforcement Learning (Advanced) or Supervised Learning (Foundational)
RL Justification	The agent (app) learns from user interactions; actions are choosing the next question; rewards are based on user success and engagement. Learns a personalised curriculum.
SL Justification	Train a model to predict the probability a user will answer a question correctly. Use this to select questions of optimal difficulty. Simpler to implement than RL.
Why Others Are Less Suitable	Unsupervised Learning could group users by skill level but cannot make dynamic decisions about what to present next.

Justification Framework for Students

Use this structure when presenting your pitch:

1. State your choice: "We propose using [ML Approach] ."

2. Explain the fit: "This is because the problem requires [characteristic, e.g., learning from interaction, classifying labeled images] ."

3. Reject an alternative: "[Alternative ML Approach] is less suitable because [reason, e.g., we don't have labeled data, the problem isn't about finding hidden groups] ."

7. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Introduction	10 min	Guiding question; Netflix scenario brainstorming
ML Approaches	25 min	Detailed presentation on 5 ML approaches (Supervised, Unsupervised, Reinforcement, Deep, Transfer)
Consultant Project	20 min	Groups analyse a problem, choose ML approach, prepare 2-minute pitch
Consolidation	5 min	Share one or two pitches; summarise key differentiators

8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions during discussions
Consultant Project	Evaluate ability to select and justify appropriate ML approach
Worksheet	Assess understanding of ML types and applications
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the key difference between supervised and unsupervised learning?
2. Give an example of a problem suited to reinforcement learning.
3. Why is transfer learning useful?
4. What type of data does supervised learning require?

9. Differentiation

Level	Strategy
Support	Provide visual aids and analogy-based explanations; pair with confident students
Challenge	Research real-world implementations; explore advanced topics (GANs, Transformers)

10. Resources

- **Presentation** (Slide Deck)
- **eBook** (A4.1.1 with examples)
- **Worksheet:** ML type matching and justification
- **Worksheet Answers**
- **Quizlet** (A4.1.1 vocabulary flashcards)

11. Summary: Key Takeaways

ML Type	Key Feature	Best For
Supervised Learning	Labelled data; maps inputs to outputs	Classification, regression, prediction
Unsupervised Learning	Unlabelled data; finds patterns	Clustering, segmentation, anomaly detection
Reinforcement Learning	Trial and error; learns from rewards	Robotics, games, optimisation
Deep Learning	Neural networks with many layers	Image recognition, NLP, speech
Transfer Learning	Reuses pre-trained knowledge	Reducing data and training time

text

KEY REMINDERS	
✔	Supervised = Labeled data, known outputs
✔	Unsupervised = Unlabeled data, hidden patterns
✔	Reinforcement = Trial and error, rewards
✔	Deep Learning = Neural networks with many layers
✔	Transfer Learning = Reusing knowledge from one task
✔	Deep Learning is a TOOL used in other approaches
✔	Transfer Learning is a STRATEGY to improve efficiency

"Machine learning is about teaching computers to learn from data—different problems require different approaches, from supervised learning for classification to reinforcement learning for interactive decision-making."